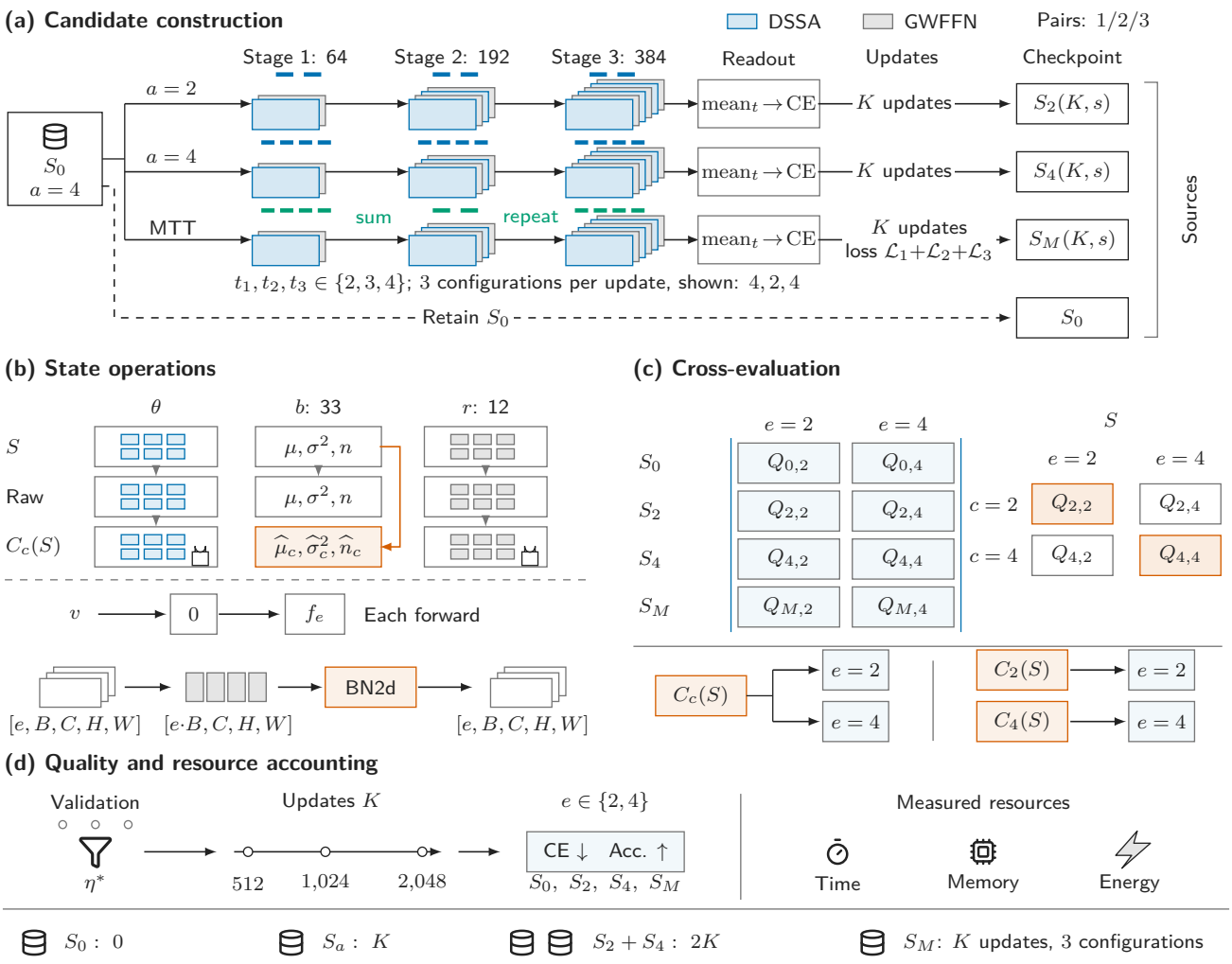

(a) Candidate construction
DSSA
GWFFN
Pairs: 1/2/3
Stage 1: 64
Stage 2: 192
Stage 3: 384
Readout
Updates
Checkpoint
$S_0$
$a = 4$
$a = 2$
$a = 4$
MTT
sum
repeat
$\text{mean}_t \to \text{CE}$
$K$ updates
$K$ updates
loss $\mathcal{L}_1+\mathcal{L}_2+\mathcal{L}_3$
$S_2(K,s)$
$S_4(K,s)$
$S_M(K,s)$
Sources
$t_1, t_2, t_3 \in \{2,3,4\}$; 3 configurations per update, shown: $4, 2, 4$
Retain $S_0$
$S_0$
(b) State operations
$\theta$
$b$: 33
$r$: 12
$S$
Raw
$C_c(S)$
$\mu, \sigma^2, n$
$\mu, \sigma^2, n$
$\widehat{\mu}_c, \widehat{\sigma}^2_c, \widehat{n}_c$
$v$
0
$f_e$
Each forward
$[e, B, C, H, W]$
$[e{\cdot}B, C, H, W]$
BN2d
$[e, B, C, H, W]$
(c) Cross-evaluation
$e = 2$
$e = 4$
$S$
$S_0$
$S_2$
$S_4$
$S_M$
$Q_{0,2}$ $Q_{0,4}$
$Q_{2,2}$ $Q_{2,4}$
$Q_{4,2}$ $Q_{4,4}$
$Q_{M,2}$ $Q_{M,4}$
$c = 2$
$c = 4$
$Q_{2,2}$ $Q_{2,4}$
$Q_{4,2}$ $Q_{4,4}$
$C_c(S)$
$e = 2$
$e = 4$
$C_2(S)$
$e = 2$
$C_4(S)$
$e = 4$
(d) Quality and resource accounting
Validation
$\eta^*$
Updates $K$
512
1,024
2,048
$e \in \{2, 4\}$
CE ↓ Acc. ↑
$S_0$, $S_2$, $S_4$, $S_M$
Measured resources
Time
Memory
Energy
$S_0$ : 0
$S_a$ : $K$
$S_2 + S_4$ : $2K$
$S_M$: $K$ updates, 3 configurations